

BISHWA PRASAD GHIMIRE

AI/ML Developer | Computer Vision & Deep Learning | BSc CSIT Student

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SUMMARY

BSc CSIT student at Tribhuvan University focused on Artificial Intelligence, Machine Learning, and Computer Vision. Experienced in building and evaluating machine learning and deep learning models using Python, PyTorch, and Scikit-learn. Developed a CNN-based Nepali handwritten character recognition system achieving 98.92% accuracy across 58 classes, including dataset preprocessing, augmentation, hyperparameter tuning, and character-wise performance analysis. Actively seeking AI/ML research and internship opportunities to apply and extend this experience.

EDUCATION

BSc in Computer Science & Information Technology (BSc CSIT)

2024 – Present

Tribhuvan University · Butwal Multiple Campus

Relevant Courses: Artificial Intelligence, Data Structures & Algorithms, Database Management Systems

TECHNICAL SKILLS

Programming: Python, SQL, C/C++

Machine Learning: Supervised Learning, Classification, Regression, Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning

Deep Learning: CNNs, Fine-Tuning, Transfer Learning, Image Classification, Model Training & Evaluation

Computer Vision: OpenCV, Image Preprocessing, Data Augmentation, Object Detection

Frameworks & Libraries: PyTorch, Scikit-learn, Transformers, NumPy, Pandas, Matplotlib, Seaborn

Tools: Git/GitHub, Jupyter, Google Colab, VS Code, FastAPI

RESEARCH EXPERIENCE

Undergraduate Researcher — Handwritten Nepali OCR / Computer Vision

2025 – Present

Independent Research · Tribhuvan University

- Investigating a **shifted-window Transformer and byte-level decoder architecture** for recognizing complex and compound characters in handwritten Nepali manuscripts.
- Exploring **Swin Transformer-based visual representations** to capture both local character structures and broader contextual information from handwritten manuscript images.
- Designing a **byte-level decoding approach** to represent and generate Unicode Nepali text while handling compound characters and variable character composition.
- Developing experimental pipelines for dataset preprocessing, augmentation, model training, evaluation, and character-level error analysis.
- Comparing Transformer-based approaches with CNN baselines to study recognition accuracy, generalization, and robustness on handwritten Nepali data.

PROJECTS

SafeVision-AI — AI-Powered Public Safety System

2026

Computer Vision · Deep Learning · YOLO

- Built a computer vision pipeline integrating multiple YOLO-based detection modules for real-time public safety monitoring, **covering violence detection, weapon detection, crowd density estimation, fire/smoke detection, and traffic analysis.**
- Fine-tuned pretrained YOLO models on task-specific datasets for each detection module and evaluated performance using precision, recall, and mAP.
- Built preprocessing and inference pipelines to handle video streams and custom datasets across the different detection tasks.
- Optimized model inference for real-time performance suitable for deployment on live video feeds.

Tech: Python, PyTorch, YOLO, OpenCV, NumPy

Nepali Handwritten Character Recognition — CNN

2026

Research Project · Computer Vision · PyTorch

- Designed CNN to classify 58 Nepali handwritten character classes with **98.92% accuracy**.
- Full pipeline: data collection, preprocessing, augmentation, train/val/test splits.
- Character-wise accuracy analysis to identify weak classes and guide improvements.
- Hyperparameter tuning: learning rate, batch size, dropout for better generalization.

Tech: Python, PyTorch, NumPy, Matplotlib, OpenCV

Movie Recommendation System

2025

Collaborative & Content-Based Filtering

- Developed a hybrid recommendation system combining **collaborative and content-based filtering** to generate personalized movie recommendations.
- Computed **cosine similarity** between users/items and movie feature representations to identify relevant recommendations.
- Preprocessed and analyzed user–movie interaction data to support personalized ranking and recommendation.

Tech: Python, Pandas, NumPy, Scikit-learn

LANGUAGES

Nepali — Native **English** — Fluent